BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SARGODHA

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021 • Paper Code: 5197 / 5192

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Q. Paper : I (Objective Type)

سوالیہ پرچہ: I (معروضی طرز)

QQ-1. 0921 — (جماعت نہم) (Session 2017-19 to 2020-22) — Inter Part-I — Sargodha Board — (معروضی) ریاضی سائنس / Mathematics (Objective) — Group 1

Q. Time Allowed: 20 minutes — PAPER CODE 5197 — Maximum Marks: 15

Q1. Mid-point of the points (2, -2) and (-2, 2) is

(A) (B)

(C) (D)

Q2. A point equidistant from the end points of a line-segment is on its ____

(A) (B)

(C) (D)

Q3. One angle on the base of an isosceles triangle is 30°. What is the measure of its vertical angle

(A) (B)

(C) (D)

Q4. Order of transpose of [[2,1],[0,1],[3,2]] is ____

(A) (B)

(C) (D)

Q5. If |2 6; 3 x| = 0, then x is equal to

(A) (B)

(C) (D)

Q6. In $\sqrt[3]{35}$ the radicand is ____

(A) (B)

(C) (D)

Q7. The conjugate of 5 + 4i is ____

(A) (B)

(C) (D)

Q8. log p - log q = ____

(A) (B)

(C) (D)

Q9. log m^n can also be written as ____

(A) (B)

(C) (D)

Q10. $a^3 + b^3$ is equal to ____

(A) (B)

(C) (D)

Q11. Factors of $3x^2 - x - 2$ are

(A) (B)

(C) (D)

Q12. Simplify $(a^2+5a-14)/(a^2-3a-18) \times (a+3)/(a-2)$

(A) (B)

(C) (D)

Q13. The square root of $x^4 + 1/x^4 + 2$ is ____

(A) (B)

(C) (D)

Q14. x = ____ is a solution of the inequality $-2 < x < 3/2$

(A) (B)

(C) (D)

Q15. If y = 2x + 1, x = 2 then y is

(A) (B)

(C) (D)

Board typos kept verbatim: 'One angle on the base of an isosceles triangle', 'Right bisector', 'In $\sqrt[3]{35}$ the radicand is'. The objective is printed in a bilingual table (Urdu + English); questions transcribed from the printed English column.

Paper : I (Essay Type)

پرچہ: I (انشائیہ طرز)

QPart I. 0921 — (**جماعت نہم**) (Session 2017-19 to 2020-22) — Inter Part-I — Sargodha Board — (**انشائیہ**) **ریاضی سائنس** / Mathematics (Essay Type) — Group 1

Q. Time Allowed: 2:10 hours — Maximum Marks: 60

QPart II. Part II — **حصہ دوم**

Note: Attempt any Three Questions. Q.9 is compulsory (8×3=24)

Board typos kept verbatim: 'if it exist', 'Answcr briefly' (Q4 header), 'matrix inversion method', 'Facorize', 'unknowns x and m', 'Answer briefly any Six parts from the followings'. Subjective parts transcribed from the printed English column; the identical Urdu column appears in the scans as printed.

GROUP – II

Q. Paper : I (Objective Type)

سوالیہ پرچہ: I (معروضی طرز)

QQ-1. 0921 — (**جماعت نہم**) (Session 2017-19 to 2020-22) — Inter Part-I — Sargodha Board — (**معروضی**) **ریاضی سائنس** / Mathematics (Objective) — Group 2

Q. Time Allowed: 20 minutes — PAPER CODE 5192 — Maximum Marks: 15

Q1. If x + [[-1,-2],[0,-1]] = [[1,0],[0,1]] then x = ____

(A) (B)

(C) (D)

Q2. The order of matrix [2 1] is

(A) (B)

(C) (D)

Q3. The value of i^9 is ____

(A) (B)

(C) (D)

Q4. Write 4^(2/3) with radical sign

(A) (B)

(C) (D)

Q5. The value of log(p/q) is ____

(A) (B)

(C) (D)

Q6. log p - log q is same as ____

(A) (B)

(C) (D)

Q7. 4x + 3y - 2 is an algebraic ____

(A) (B)

(C) (D)

Q8. The factors of x^2 - 5x + 6 are ____

(A) (B)

(C) (D)

Q9. H.C.F of x - 2 and x^2 + x - 6 is ____

(A) (B)

(C) (D)

Q10. What should be added to complete the square of x^4 + 64 ____ ?

(A) (B)

(C) (D)

Q11. If x is no larger than 10, then ____

(A) (B)

(C) (D)

Q12. If (x - 1, y + 1) = (0, 0) then (x, y) is ____

(A) (B)

(C) (D)

Q13. Distance between points (0, 0) and (1, 1) is ____

(A) (B)

(C) (D)

Q14. A triangle having two sides congruent is called ____

(A) (B)

(C) (D)

Q15. The diagonals of a parallelogram ____ each other.

(A) (B)

(C) (D)

Board typos kept verbatim: 'isosceles' options list 'Isosceles' as 'Isosceles' (printed spelling), 'What should be added to complete the square of $x^4 + 64$', 'algebraic' printed 'algebric'. The objective is printed in a bilingual table (Urdu + English); questions transcribed from the printed English column.

Paper : I (Essay Type)

پرچہ: I (انشائیہ طرز)

QPart I. 0921 — (جماعت نہم) (Session 2017-19 to 2020-22) — Inter Part-I — Sargodha Board — (انشائیہ) ریاضی سائنس / Mathematics (Essay Type) — Group 2

Q. Time Allowed: 2:10 hours — Maximum Marks: 60

QPart II. Part II — حصہ دوم

Note: Attempt any Three Questions. Q.9 is compulsory (8×3=24)

Board typos kept verbatim: 'Find the unknown value of the figure', 'Find the Area of a figure', 'Define In Centere', 'Answer briefly any Six parts from the followings', '4x + 2y = 8' (Q5a as printed), 'matrix inversion method'. Subjective parts transcribed from the printed English column; the identical Urdu column appears in the scans as printed.